

HAYDEN FU

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EDUCATION

Cornell University, College of Arts & Sciences

Expected May 2028

Computer Science and Statistics | GPA: 3.78

Relevant Coursework: Algorithms, Computer Architecture, Machine Learning, Data Structures and Functional Programming, Object-Oriented Programming, Discrete Mathematics, Probability Models and Inference

EXPERIENCE

Cornell University, Civil Engineering | *Student Researcher*

Jun 2026 - Present

- Built a Python scientific-computing pipeline integrating satellite observations, numerical simulations, and geospatial datasets to estimate unobserved irrigation events across multiple years.
- Automated experiments that generated candidate inputs, launched VIC simulations on Cornell's SLURM cluster, parsed NetCDF outputs, and scored results against observations.
- Implemented adaptive probabilistic sampling and particle-weighting algorithms to refine candidate irrigation schedules and estimate posterior water use.
- Built reusable preprocessing and validation modules for spatial matching, time-series alignment, rainfall filtering, and model-output extraction; evaluated ML bias correction using temporal and geographic holdouts.

LLM Forecasting Research | *Researcher*

2026

- Built a Python forecasting pipeline combining prediction-market data, historical order books, timestamped news, and LLM inference to study long-horizon forecast behavior.
- Designed time-aware evidence retrieval using dense embeddings and cross-encoder reranking to select relevant historical news while preventing future-information leakage.
- Built experimental workflows for Qwen and Gemma under multiple context and prior-forecast conditions; developed metrics for anchoring, forecast instability, and directional updates.

Cornell GeoData | *Student Researcher*

2025 - 2026

- Developed Python tools to extract and process HRRR weather-model data and surface-station observations for lake-effect snow analysis around Cayuga Lake.
- Implemented geospatial sampling and features for temperature differentials and wind shear; reduced redundant spatial lookups and organized extraction into reusable functions.

PROJECTS

Six Hops | BigRed//Hacks 2026 | Capital One & Solana Track Winner

Next.js, TypeScript, PostgreSQL, Solana

- Co-developed a full-stack referral search platform using Next.js, TypeScript, and PostgreSQL, integrating iMessage workflows and automated Solana devnet payouts.
- Built interactive referral-network visualizations and multi-hop search features to trace connections and examine search propagation.

OCaml Chess

OCaml

- Built an approximately 869-line chess project using immutable data structures, recursion, and modular abstractions; implemented board-state transitions and move-validation logic.

SKILLS

Languages: Python, TypeScript, Java, C/C++, OCaml, JavaScript, R, SQL

Libraries and Frameworks: Next.js, React, Node.js, Express, Tailwind CSS, pandas, NumPy, xarray, scikit-learn

Tools and Platforms: Git, PostgreSQL, Linux, SLURM, OpenMPI, Jupyter, Azure AI Foundry, NetCDF